Muqeeth Mohammed

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Education

- **University of North Carolina - Chapel Hill** — Chapel Hill, NC
  Masters of Science in Computer Science — *Aug 2021 - May 2023*
  - **Research Interests**: Natural Language Processing, Transfer Learning, Machine Learning, Optimization
- **Indian Institute of Technology Madras** — Chennai, India
  Bachelor of Technology (Honors) in Electrical Engineering — GPA:9.36/10.0 (Rank 8/60) — *Jul 2016 – May 2020*
  - **Coursework**: Machine Learning, Non-Linear Optimization, Probabilistic Graphical Models
  - **Thesis**: Estimation of relatedness between courses in Workflow

Experience

- **University of North Carolina - Chapel Hill** — Chapel Hill, NC
  Graduate Student Researcher — Advisor: Prof. Colin Raffel — *Aug 2021 - May 2023*
  - Working on developing algorithms that can overcome the issue of not having well-defined gradients in models that use conditional computation, for example, Mixture of experts (MoEs)
- **Indian Institute of Technology, Delhi** — Delhi, India
  Research Assistant — Advisors: Prof. Mausam, Prof. Soumen Chakrabarti — *Jul 2020 - Jul 2021*
  - Worked on Multilingual Open Information Extraction with focus on low resource languages
  - Also worked on a unified labeling architecture for structured prediction tasks that include syntax, semantics, information extraction, and sentiments in NLP which is 3x faster than state-of-the-art achieving nearly the same accuracy
- **Senior Thesis** — Chennai, India
  Undergrad Student — Advisors: Prof. Sutanu Chakraborti, Prof. Anil Prabhakar — *Jan 2020 - May 2020*
  - Designed a retrieval system to retrieve relevant courses to a given query course from our university courses
  - Used explicit semantic analysis with the addition of phrases to achieve 0.868 nDCG score compared to a neural model baseline of 0.735. Implemented the idea of retrofitting to user feedback to further improve the score to 0.92
- **Flipkart** — Bengaluru, India
  Summer Intern — *May 2019 - July 2019*
  - Built an unsupervised classification model to classify opinion-aspect pairs from Flipkart review corpus for shoes into categories. Achieved an accuracy of 76 % on human-annotated test set
  - Implemented decision trees using features from the review sentence's dependency parse to identify the categories like Quality, Comfort, etc. implicitly or explicitly spoken in a review with an F1 score close to 86% for each category

Publications

- **Alignment-Augmented Consistent Translation for Multilingual Open Information Extraction (ACL, 2022)**
  Keshav Kolluru, **Mohammed Muqeeth***, Shubham Mittal, Soumen Chakrabarti, Mausam*

Selected Projects

- **Multidomain Image Classification'21**: Modeled image classificatiom from diverse domains using MOEs. This modeling reduced the negative transfer between images from multiple domains to improve the final classification accuracy
- **Movie Recommender System'19**: Built an ensemble model of collaborative filtering using the neighborhood method and SVR regression to predict user rating for a movie in the MovieLens dataset.
- **InterIIT Mobile App'17**: Designed features like live streaming, live score, and a gallery page for the InterIIT games mobile app using Ionic. Deployed the app successfully onto the play store with more than 5k downloads and a 4.3/5 rating

MISCELLANEOUS

- Received Masters Merit Fellowship for the first year at UNC Chapel Hill
- Recevied RamShriram Merit Scholarship at IIT Madras for maintaining 8.0+ GPA every year
- Reviewer: ACL 2022 (ACL Rolling Review)

Technologies

- Python, PyTorch, Sklearn, Numpy, JavaScript, LaTeX